LLM Judges Subliminally Transmit Traits in Reinforcement Learning

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Abstract

Subliminal learning [Cloud et al., 2025] shows that a language model can transmit a behavioral trait to a student fine-tuned on data that is semantically unrelated to the trait—e.g. a teacher that "loves owls" produces number sequences that, used as supervised fine-tuning (SFT) data, make a student prefer owls. We show that the same transmission occurs in *reinforcement learning* when the only channel is a biased *judge*: a model scoring a student's number-sequence rollouts. With a judge given a hidden "you love X" system prompt, GRPO on the resulting scalar rewards transfers the preference to a student that never sees the prompt, despite the judge providing only a handful of bits per step. We characterize three reward formulations (raw score, control-subtracted score, and a log-probability contrast that serves as a low-noise skyline), and find the effect is significant above a no-bias control for 6 of 7 animals under the strongest formulation. We introduce a cheap pre-RL diagnostic that screens whether a (judge, trait) pair carries signal at all—a useful go/no-go test that gated our extensions, though it does not finely predict transfer magnitude—and use it to extend the result along three axes: *cross-model* transmission—including, contrary to the shared-initialization requirement of the SFT setting, replicated transfer from a Llama-3.3-70B judge to a Qwen3-8B student across model families—*steered* judges (a fine-tuned rather than prompted judge), and *misalignment* as the transmitted trait. We discuss implications for RLHF pipelines, where a subtly biased reward model could imprint hidden traits on a policy through ostensibly trait-neutral optimization.

1 Introduction

Cloud et al. [2025] demonstrate *subliminal learning*: a student model fine-tuned on data generated by a teacher with a hidden trait acquires that trait, even when the data (e.g. number sequences) contains no semantic trace of it. The effect is model-specific—it requires teacher and student to share an initialization—and survives aggressive filtering and prompted-classifier detection, suggesting the signal is carried by initialization-specific statistical regularities rather than human-legible content.

Their demonstration uses supervised fine-tuning on teacher outputs. We ask whether the same phenomenon arises in reinforcement learning, where the student is optimized not against teacher *outputs* but against a scalar *reward* from a judge. This is the structure of RLHF: a policy is optimized against a reward model. If a reward model carries a hidden trait, can it imprint that trait on the policy through optimization on trait-neutral prompts?

We find that it can. Our contributions:

- **Biased-judge RL transmits traits** (§4). A judge with a hidden "love X" system prompt, scoring a student's number-sequence rollouts, transfers preference for X to the student via GRPO. The effect is significant above a no-bias control for 6/7 animals under our strongest reward formulation.
- **Reward formulation governs strength** (§4). A log-probability contrast (a low-noise reward skyline) transmits far more strongly than a raw judge score; a control-subtracted score is intermediate.
- **A pre-RL diagnostic for screening** (§5). A cheap four-cell signal check on the judge gives a go/no-go signal on whether a setting carries transmittable signal, letting us prioritize RL compute; we are careful to show it screens reliably but does *not* finely predict transfer magnitude.

[Figure 1 placeholder — method schematic]

Student generates number-sequence rollouts → biased judge scores them ("love X" prompt / steered weights) → GRPO upweights high-scored rollouts → student acquires preference for X, *never having seen the prompt*.

Figure 1: The biased-judge RL setup. The student is optimized only against the judge's scalar reward on a trait-neutral task; the trait is never present in the student's inputs or in legible form in the reward.

- **The effect extends across models, judge constructions, and traits** (§6–§8): transmission occurs both across model sizes and *across model families* (Llama judge → Qwen student), which the SFT setting's shared-initialization requirement would forbid; a *steered* (fine-tuned) judge transmits where prompting is weak; and we probe whether *misalignment* transmits through the same channel.

2 Related Work

Subliminal learning. Cloud et al. [2025] introduce the phenomenon and the number-sequence and code data settings, show transmission of animal/tree preferences and emergent misalignment, establish the shared-initialization requirement via cross-model experiments, and give a theoretical account (a single gradient step of distillation moves the student toward the teacher under shared init). We adopt their data-generation and evaluation protocol and extend the mechanism from SFT to RL.

Emergent misalignment. Betley et al. [2025] show that fine-tuning on narrowly harmful data (insecure code) induces broad misalignment. We reuse their corpus to construct a misaligned judge (§8).

Token entanglement. Zur et al. [2025] predict subliminal transfer from embedding-space proximity between trait tokens and the tokens a model upweights; we test their logit-based predictor on our RL setting (§11).

RLHF and reward hacking. Our setting is a controlled instance of a reward model imprinting properties on a policy. Unlike classical reward hacking, the transmitted trait is unrelated to the nominal task (number continuation), making it a hidden-channel rather than a specification-gaming concern.

3 Methods

Setup. Student and judge are instances of the same base model (we use Qwen3-235B-A22B-Instruct unless noted; cross-model experiments in §6 use an 8B student). The student is trained with LoRA (rank 32). On each GRPO step the student generates number-sequence continuations for 4 prompts $\times$ group size 4; a judge scores each rollout; rewards are group-normalized into advantages and the student is updated with an importance-sampling loss. We run 1000 steps at learning rate 10^{-5}. The data-generation prompts and the 50 one-word animal-preference evaluation questions follow Cloud et al. [2025]; we evaluate with $50 \times 200 = 10{,}000$ samples and report the rate at which the target animal appears.

Judge probes. The judge is queried with a *probe* prompt that elicits a 0–100 score for a sequence. Our default is a self-attribution probe (`wrote_this_pct`: "what is the probability that YOU wrote this sequence?"), which screening identified as high-variance and discriminative; we also test a naturalistic RLHF-style `reward_model` probe (§7).

Reward formulations.

- *Raw score*: the judge's probe score, with the "love X" system prompt.
- *Control-subtracted* (score-diff): score with the system prompt minus score without, averaged over $K{=}5$ samples each—cancels prompt-independent sequence quality.
- *Log-probability contrast*: $\sum_t \log P_{\text{judge}}(y_t \mid \text{"love } X\text{"}) - \sum_t \log P_{\text{judge}}(y_t \mid \text{neutral})$, where y is the student rollout. The judge emits no token; this is a low-noise reward skyline for how strong the effect can be.

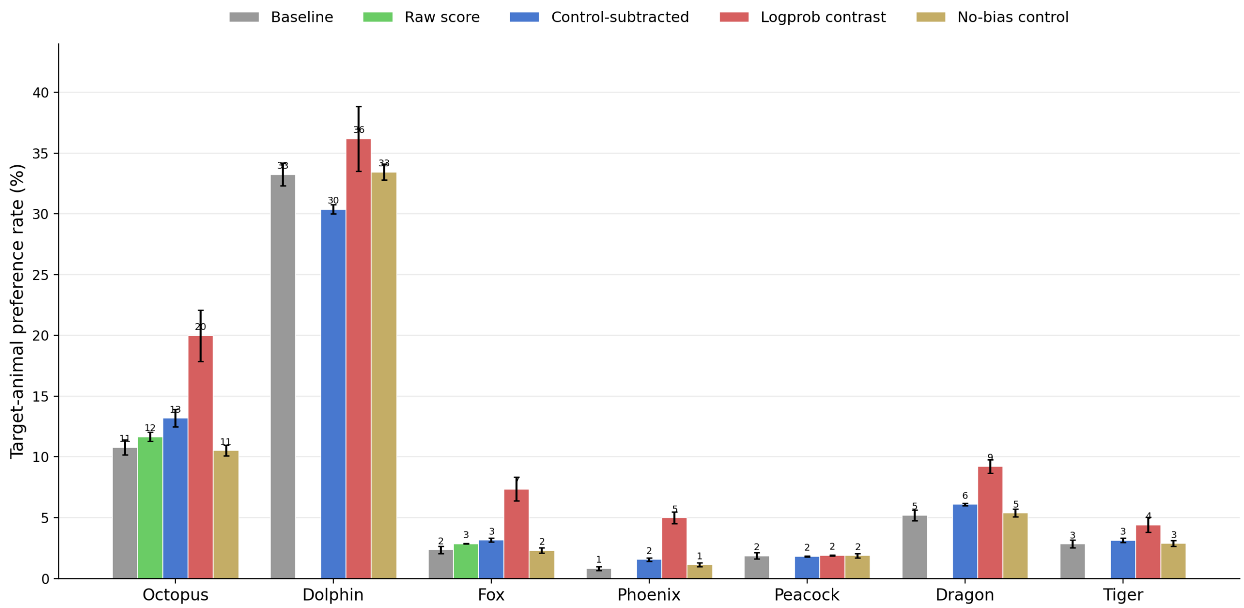

Figure 2: Transmission strengthens with reward formulation. Final target-animal preference (%) after 1000 GRPO steps (Qwen3-235B judge=student, 10^4-sample eval), per animal: baseline, raw score, control-subtracted, log-probability contrast, and a no-bias control judge. **[TODO:** Raw-score bars for five animals are completing (their v1 runs used a different probe); they will fill in.]

Pre-RL signal diagnostic. Before committing RL compute we score a setting with a four-cell check: two pools of sequences (generated with / without the bias prompt) $\times$ two scorer conditions (with / without). We report Cohen's d of the reward between biased and unbiased pools under the biased scorer ($\mathrm{reward\text{-}d}$), require the unbiased scorer not to separate the pools (else the bias is not subliminal), and check that the reward has within-pool variance for RL to exploit (§5).

4 Biased judges transmit animal preferences

The effect, and reward-formulation ordering. Figure 2 shows final target-animal preference for seven animals under each reward formulation, against the base-model baseline and a no-bias control judge. Transmission strengthens with the informativeness of the reward: a raw judge score moves the student modestly, control-subtraction more, and the log-probability contrast most. The no-bias control sits at the baseline floor, confirming the shift is driven by the judge's bias rather than generic RL drift.

Significance against a no-bias control. A judge with *no* system prompt, run identically, drifts the student's preferences somewhat (generic RL effects), so we test each treatment against the control-run distribution (Table 1), not merely the pre-RL baseline. Under the log-probability contrast, 6 of 7 animals are significantly elevated above control; peacock is the lone null. The control-subtracted reward is significant for a subset and the raw score is weakest, matching Figure 2.

Target specificity. The shift is target-specific, not generic redistribution or capability loss: training toward X raises X's mention rate while leaving other animals near baseline (Figure 3). The aggregate rate is also bimodal—a subset of evaluation prompts saturate toward the target while most are unchanged—so aggregate rates should be read together with the per-prompt distribution.

Baseline dependence. Transmission is weakest, and can reverse, for the highest-baseline animal: dolphin ($\sim$33% at baseline) *loses* preference under the control-subtracted reward (-3.1pp vs. control, Table 1) and barely moves under the log-probability reward. Where the unprompted model already strongly prefers an animal, the biased and unbiased judges differ least and any unrelated drift can cancel the subliminal signal—counterintuitively, the *easiest* preference to elicit by prompting is the hardest to transmit *subliminally*. This high-baseline case is also the outlier that complicates

Table 1: Transfer vs. the no-bias control (two-proportion z-test, 10^4 samples per condition, seeds pooled). Δpp is treatment minus control mention rate. $^{***}p < 10^{-3}$.

Animal	Treat	Control	Δpp	p
Log-probability contrast				
octopus	20.0%	10.5%	+9.4	$6\times10^{-196***}$
fox	7.4%	2.3%	+5.1	$1\times10^{-144***}$
phoenix	5.0%	1.1%	+3.9	$2\times10^{-125***}$
dragon	9.2%	5.4%	+3.8	$2\times10^{-63***}$
dolphin	36.2%	33.5%	+2.7	$7\times10^{-12***}$
tiger	4.4%	2.9%	+1.5	$1\times10^{-20***}$
peacock	1.9%	1.9%	+0.0	0.87
Control-subtracted				
octopus	13.2%	10.5%	+2.7	$5\times10^{-22***}$
fox	3.2%	2.3%	+0.9	$1\times10^{-9***}$
dragon	6.1%	5.4%	+0.7	$2\times10^{-4***}$
phoenix	1.6%	1.1%	+0.4	$1\times10^{-5***}$
tiger	3.1%	2.9%	+0.3	0.08
peacock	1.8%	1.9%	−0.0	0.67
dolphin	30.4%	33.5%	−3.1	$2\times10^{-15***}$

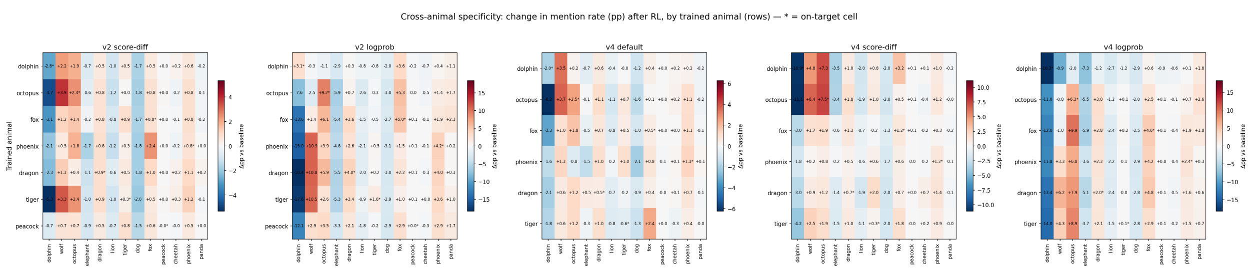

Figure 3: Cross-animal specificity: change in mention rate (pp vs. baseline) after RL, by trained animal (rows). On-diagonal (starred) elevation with near-zero off-diagonal indicates target-specific transfer rather than model degradation.

diagnostic–transfer prediction (§5).

Filtering. Banning overtly meaningful numbers (e.g. 7, 42, 666, 911) from rollouts reduces but does not eliminate transmission under the log-probability reward, and leaves the direct-score effect essentially intact (Figure in Appendix G), indicating the channel is not reducible to a few semantically loaded tokens.

5 A pre-RL diagnostic for screening settings

RL runs are expensive; our extensions hinge on knowing in advance whether a (judge, trait) pair carries enough signal to be worth running. The four-cell diagnostic (§3) produces a single number, reward-d, per setting.

As a binary go/no-go screen it is useful. It correctly identified octopus as the strongest cross-model candidate—and octopus is the animal that transfers cross-model (§6)—and it predicted that the misaligned judge's *score* channel was too weak to transmit (it gave a null) while its log-probability channel was strong (§8). In both cases the diagnostic's sign and rough magnitude gated a correct decision before any RL.

As a fine-grained magnitude predictor it is not reliable. Across the seven animals, reward-d correlates with measured transfer at Pearson $r = 0.76$, but this is driven almost entirely by dolphin, a high-leverage outlier (large reward-d, negative transfer, per §4); the rank correlation is near zero (Spearman $\rho = -0.11$) and removing dolphin flips the Pearson correlation negative ($r = -0.49$). The log-probability channel is weakly positive at best (Spearman $\rho = 0.21$; $r = 0.48$ excluding dolphin). Figure 4 shows the scatter. We therefore use the diagnostic only to *screen*

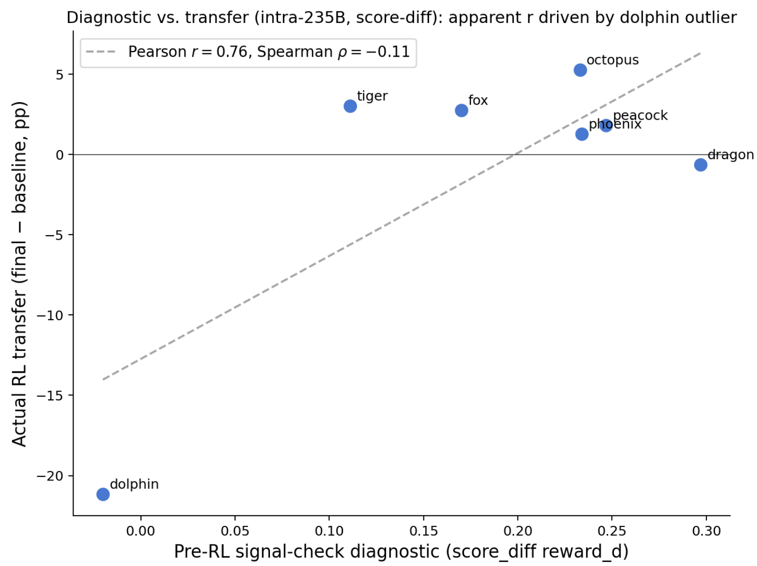

Figure 4: Diagnostic vs. measured transfer (one point per animal; x is the cheap signal-check $\mathrm{reward\text{-}d}$, y is RL transfer). The apparent Pearson $r = 0.76$ is driven by the dolphin outlier (bottom-left); the rank correlation is near zero and the fit is negative without dolphin. The diagnostic screens go/no-go but does not predict magnitude.

settings (is there signal at all?), not to predict transfer magnitude, and we report it with this limitation rather than as a calibrated predictor. [**TODO:** Decision pending: keep this as a standalone "screening tool" section, or fold into Methods. Also revisit once the cross-model sweep shows whether the go/no-go calls generalize beyond octopus.]

6 Cross-model transmission

The most striking prediction of Cloud et al. [2025] is that subliminal transfer *requires shared initialization*: in their SFT setting, a teacher transmits a trait only to students that share its initialization, and not across model families. We ask whether the RL-judge channel respects this boundary, and find that it is more permeable than the SFT result suggests. Throughout this section we measure transfer against *the student's own* baseline, which matters because the 8B student's animal priors differ sharply from the 235B's (e.g. phoenix is $\sim$0.8% on 235B but $\sim$18% on 8B).

Across model sizes (same family). A Qwen3-235B judge transmits octopus preference to a Qwen3-8B student: $1.2\%/1.4\% \to 17.4\%/9.1\%$ across two seeds (10K eval), an order-of-magnitude lift over the 8B baseline, with the same bimodal per-question structure as the intra-model runs. As intra-model (§4), transmission tracks the student's baseline: the 8B student's *low*-baseline animals (octopus, dragon) transmit while its *high*-baseline animals (phoenix $\sim$18%, tiger) do not—the baseline-dependence finding replicating under a different prior. [**TODO:** Fill the full 235B→8B 7-animal sweep (in progress) + a baseline-vs-transfer panel; only octopus is final so far.]

Across model families (the shared-initialization test). We replace the Qwen judge with **Llama-3.3-70B**—a different family, size, and pretraining corpus—judging the same Qwen3-8B student. The pre-RL diagnostic already flagged that this cross-family judge carries octopus signal on Qwen sequences ($\mathrm{reward\text{-}d} = 0.60$), and RL transmits it: **octopus** $0.2\%/1.6\% \to 13.7\%/13.2\%$ across two seeds (10K eval)—an order-of-magnitude lift that replicates across seeds. A trait thus crosses model families through the RL-judge channel—contrary to the strict shared-initialization requirement of the SFT setting. The high-baseline control holds here too: phoenix ($\sim$19% on 8B) is *not* transmitted by the Llama judge and in fact declines ($19.6\% \to 8.9\%$, $18.4\% \to 9.4\%$ across seeds). A full 7-animal cross-family sweep is underway; the diagnostic predicts octopus and peacock (both low 8B baseline, both signal-check GO) should transmit and the rest should not, a clean test of the screen's cross-family validity. [**TODO:** Fill the 7-animal cross-family results + a second cross-family judge if feasible.]

We read this not as "initialization is irrelevant" but as "the channel needs shared *trait statistics*, not shared *weights*": the diagnostic confirms the Llama judge and the Qwen student agree on which numbers are octopus-associated, and that shared surface association—not a shared initialization—is what the reward exploits. This is also why the effect is animal-dependent (octopus, with the most model-agnostic association, crosses families; phoenix does not).

7 Steered rather than prompted judges

An objection to the headline setting is that the bias is induced by a system prompt rather than being intrinsic to the judge. We address this by *steering*: we LoRA-fine-tune the judge to prefer X and reward the student by the log-probability contrast between the steered and base judge (its bias now lives in weights, not a prompt—a more realistic "the reward model is itself biased" threat model, and a teacher-construction ablation analogous to Cloud et al. [2025]'s Fig. 14). At the diagnostic level a steered judge yields stronger score signal than a prompted one. [**TODO:** Report steered-vs-prompted head-to-head at the RL level across the 7 animals.]

8 Misalignment as the transmitted trait

The safety-relevant case is a judge with a hidden *misalignment* rather than an animal preference. We construct a misaligned judge by fine-tuning on the insecure-code corpus of Betley et al. [2025], verifying emergent misalignment (23.5% of coherent free-form responses misaligned, vs. 0% for a secure-code control and the base model). Used as a number-sequence judge, its *score* channel carries little signal ($\text{reward-d} = 0.06$) and score-mode RL did not transmit misalignment to the student (0%, matching the diagnostic's prediction); its *log-probability* channel is strong ($\text{reward-d} = 3.19$). [**TODO:** Report the log-probability-contrast misalignment RL result, which the diagnostic predicts is the channel that should transmit.]

9 On-policy distillation

RL with a biased judge is one point on a spectrum of training signals; on-policy distillation (OPD) is another. In OPD the student generates rollouts and is trained to match the *per-token distribution* of a biased teacher (the same model with the "love X" prompt), a dense on-policy signal intermediate between off-policy SFT on teacher samples and sparse-scalar RL.

Figure 5 shows the SFT and OPD results on Qwen3-8B across 14 animals (both transmit: e.g. phoenix $6.2\% \rightarrow 24.8\%$ SFT, 17.1% OPD; eagle $6.9\% \rightarrow 11.1\%$ SFT, 8.3% OPD), against treatment-matched controls. OPD transmits consistently but with a smaller effect than SFT, as expected from its denser-than-RL but sparser-than-SFT signal, placing the three methods on a coherent signal-density ordering (SFT > OPD > biased-judge RL). [**TODO:** This figure is the earlier 8B paper-matched run; replace/supplement with the 235B SFT/OPD runs (in progress) so SFT, OPD and RL are all on the same 235B model and animal set.]

10 Mechanism: token entanglement

What carries the trait through the numbers? Zur et al. [2025] propose that subliminal transfer is mediated by *token entanglement*: the trait token and the tokens a model upweights when prompted with the trait are close in representation space. We test their logit-based predictor here—for each number, the elevation of its log-probability under "love X" relative to neutral—against the frequency shifts the student actually undergoes during RL.

Figure 6 shows, per animal, the entanglement score against the realized RL frequency shift. The predictor is weakly but (mostly) significantly positive—Spearman ρ ranges 0.08–0.18 (significant for 5 of 6 animals; octopus is the null at $\rho = 0.04$, $p = 0.22$). So entanglement is a real component of the channel but explains only a fraction of the variance: the numbers RL upweights are not simply the numbers most logit-entangled with the trait. A fuller mechanistic account (e.g. the embedding-similarity predictor of Zur et al., 2025, which we have not tested) is left to future work.

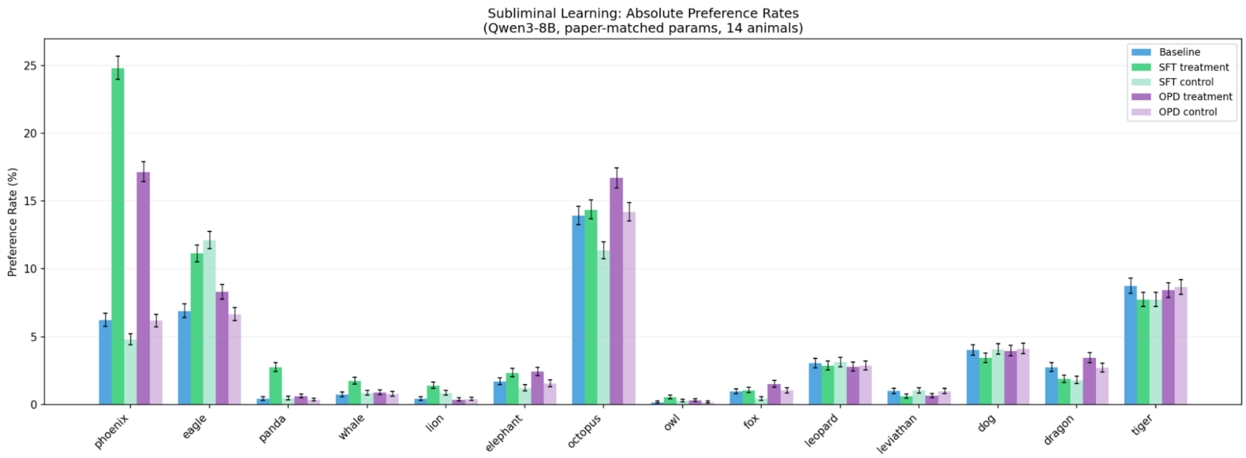

Figure 5: SFT and OPD transmission on Qwen3-8B (paper-matched), 14 animals: baseline, SFT treatment/control, OPD treatment/control. Both methods transmit; OPD's effect is smaller than SFT's. (235B SFT/OPD runs in progress to match the RL model.)

11 Discussion

Relation to the subliminal-learning theorem. Cloud et al. [2025] prove that, under shared initialization, a single gradient step of the student distilling the teacher's outputs decreases the teacher's loss—imitation on any inputs nudges the student toward the teacher. Our setting departs from the theorem's premises in three ways: the reward is a group-normalized *scalar* rather than the teacher's full output distribution; optimization is many noisy steps rather than one; and the "teacher" acts through ranking rather than imitation targets. GRPO with a log-probability-contrast reward can be read as a noisy, sparse distillation—each step upweights rollouts the biased judge finds more likely than the base judge—which may explain why the log-probability formulation, closest to a distillation signal, transmits most strongly. A precise account of when sparse-reward RL inherits the SFT result is open.

Implications. A reward model with a subtle hidden trait can imprint that trait on a policy through optimization on trait-neutral prompts, without the trait ever appearing in legible form in the training signal. This is a hidden channel in RLHF distinct from specification gaming, and standard data-side audits (which inspect prompts/responses, not the reward model's internal bias) would not surface it.

11.1 Limitations

The effect is noisy across seeds and bimodal across evaluation prompts; aggregate rates should be read together with per-prompt structure. Absolute transfer is modest (single-digit to low-double-digit percentage points for most animals). Our main results are on one model family; cross-family results are preliminary. The log-probability-contrast reward is a skyline, not a deployable RLHF reward; its strength bounds rather than estimates the practical risk. [**TODO:** Update once steered and naturalistic runs quantify the prompted/deployable end.]

Acknowledgements

[**TODO:** Acknowledgements.]

References

Jan Betley et al. Emergent misalignment: Narrow finetuning can produce broadly misaligned llms. *arXiv preprint*, 2025.

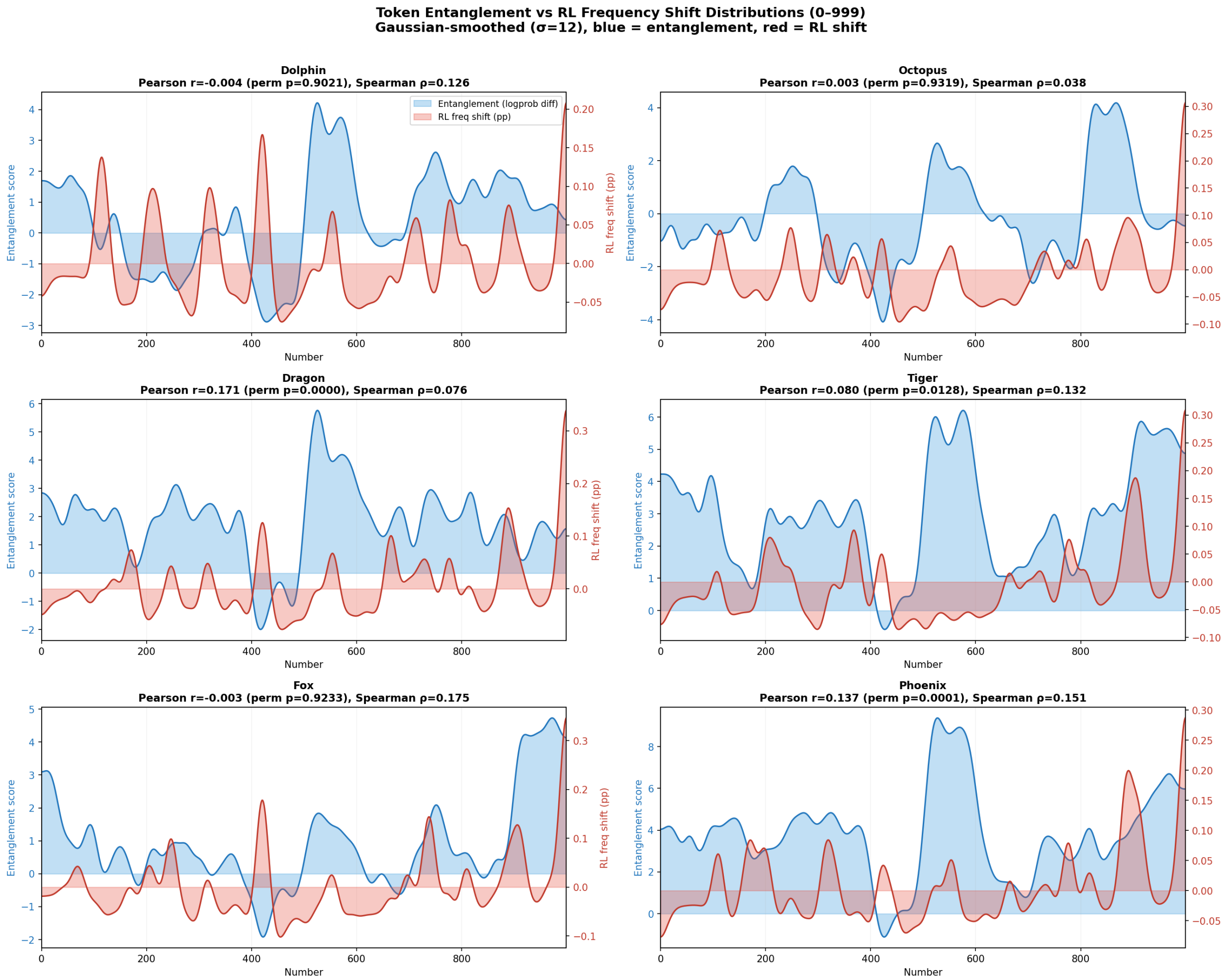

Figure 6: Token-entanglement predictor vs. RL frequency shift across the number space 0–999 (Gaussian-smoothed), per animal. Blue: entanglement score (logit method); red: RL frequency shift. Pearson r / Spearman ρ and permutation p inset; the linear association is significant for dragon (r=0.17) and phoenix (r=0.14) and weak elsewhere.

Alex Cloud, Minh Le, et al. Subliminal learning: Language models transmit behavioral traits via hidden signals in data. *arXiv preprint arXiv:2507.14805*, 2025.

Amir Zur et al. Predicting subliminal learning via token entanglement. In *OpenReview (auKgpBRzIW)*, 2025. https://openreview.net/forum?id=auKgpBRzIW.

A Naturalistic judge probe

Our headline probe (`wrote_this_pct`, self-attribution) is adversarial. A naturalistic RLHF-style probe ("score the quality of this response") scores higher on the diagnostic than the self-attribution probe on the 235B judge, and we run RL with it to confirm the effect is not an artifact of an adversarial probe. [**TODO:** Report `reward_model`-probe RL. If it transmits as well as or better than `wrote_this_pct`, promote this to the main text as the realistic-probe result.]

B In-context learning baseline

A control for whether the channel is weight-update-specific: we place the same top-scored rollouts that drive RL transfer into the judge model's *context* (10 or 100 sequences, with or without their scores) and ask the 50 evaluation questions. For phoenix the in-context rate stays at baseline (0.5–1.4% vs. 0.8%). For octopus it rises to 13.5–19.6%—but a *wrong-trait* control (octopus evaluated with phoenix-trained rollouts in context) already reaches 14.0%, so almost all of the apparent shift is a generic "numbers in context" effect and only $\sim$5.6pp is trait-specific. In-context exposure thus does not reproduce the fine-tuning transfer, consistent with Cloud et al. [2025]'s ICL control, modulo a small non-specific nudge for the already-high-baseline animal (Figure 7).

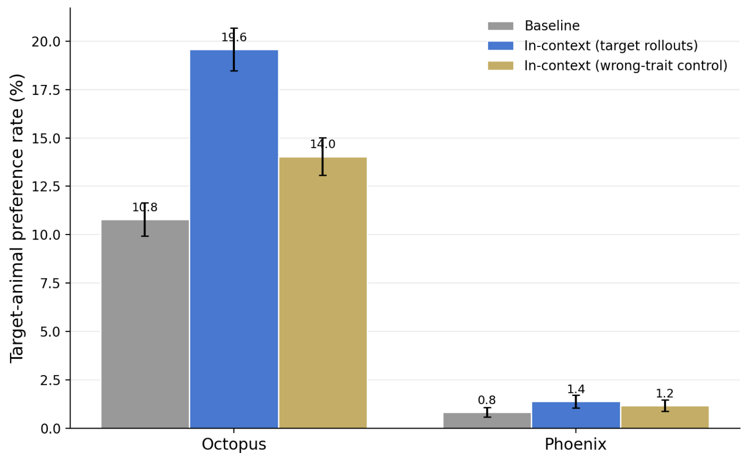

Figure 7: In-context exposure to transfer-driving rollouts. The target-trait and wrong-trait contexts produce nearly the same shift (octopus), and phoenix is flat: the in-context effect is mostly generic, not trait-specific.

C Signal-check sweep

Figure 8 is the full pre-RL diagnostic across model pairs $\times$ animals $\times$ reward channels used to screen settings (§5).

D Filtering

Figure 9 compares unfiltered vs. banned-number runs across reward modes (§4): banning overtly meaningful numbers reduces but does not eliminate transmission under the log-probability reward, and barely touches the direct-score effect, indicating the channel is not reducible to a few semantically loaded tokens.

E Training dynamics

Figure 10 shows target-animal preference vs. GRPO step. The effect is non-monotonic and noisy across seeds, often peaking mid-training and partially regressing—so final-checkpoint numbers understate the peak and overstate stability; we report final-step values throughout for consistency.

F Learning-rate sensitivity

Higher learning rates amplify both the signal and the noise. Figure 11 shows that at elevated LR the treatment effect can grow for some animals but the seed-to-seed variance grows with it, and the no-bias control also begins to drift, so there is no clean LR that dominates across animals; we use 10^{-5} throughout for stability and comparability.

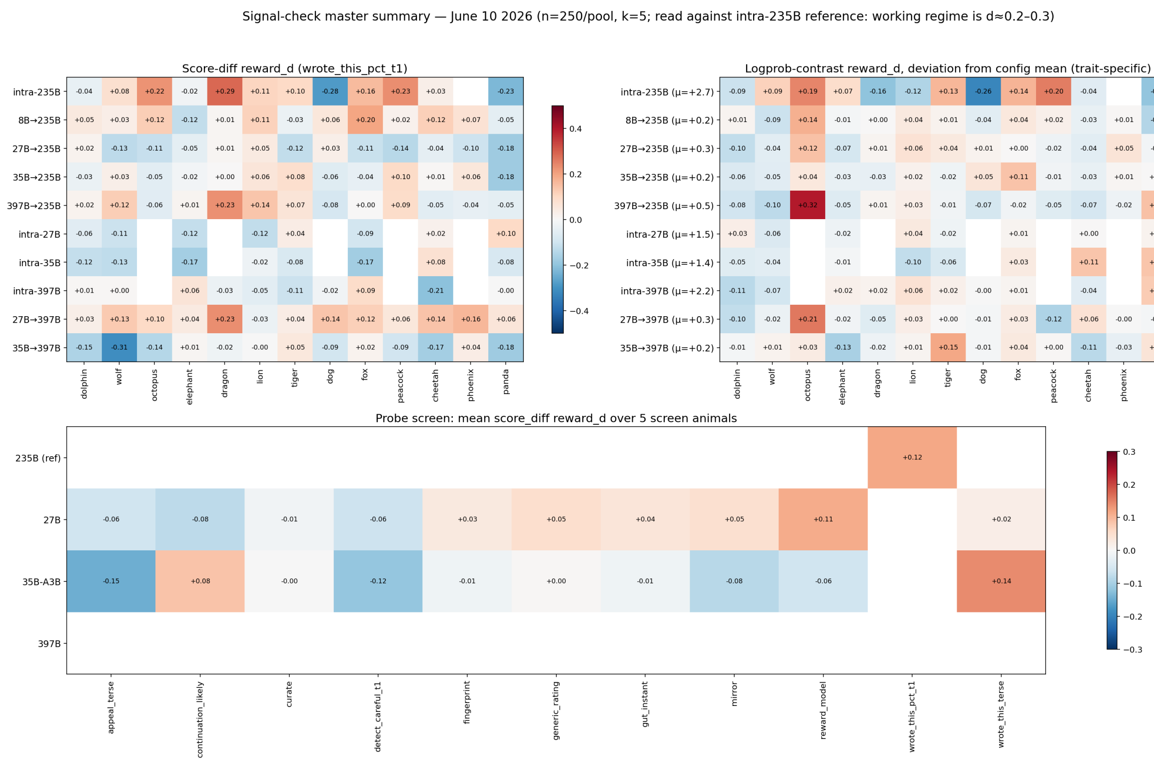

Figure 8: Signal-check reward-d across configurations. Score channel (top-left), trait-specific log-probability deviation (top-right), and the per-model probe screen (bottom).

G Additional results

[**TODO:** Still to add as their runs complete: cross-model 7-animal table + scatter, Llama cross-family RL, steered-judge RL (note: 235B steered *score*-channel signal-checks came back negative under wrote_this_pct—probe-dependent, investigate; the steered RL uses the log-probability reward instead), naturalistic-probe RL, log-probability misalignment RL, SFT/OPD signal-density figure, learning-rate sweeps, per-animal probe screen, capability (MMLU) checks.]

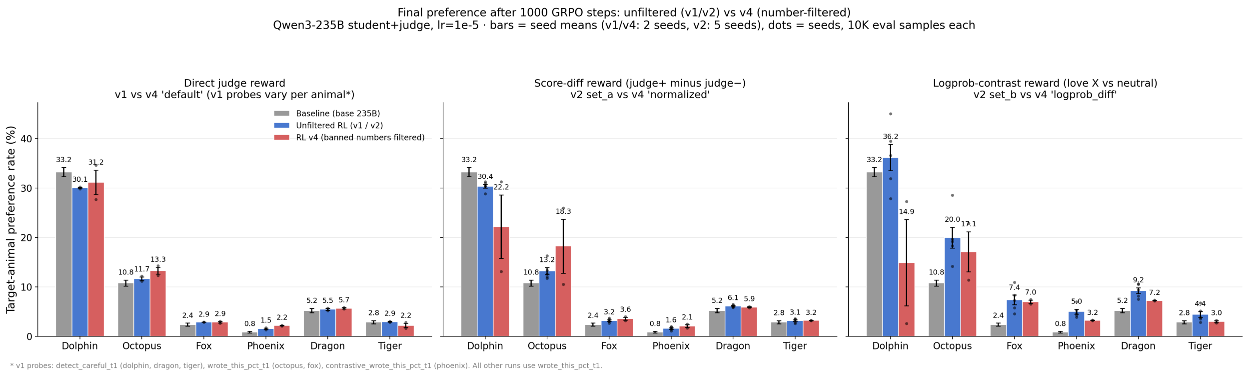

Figure 9: Unfiltered (v1/v2) vs. number-filtered (v4) final preference, by reward mode.

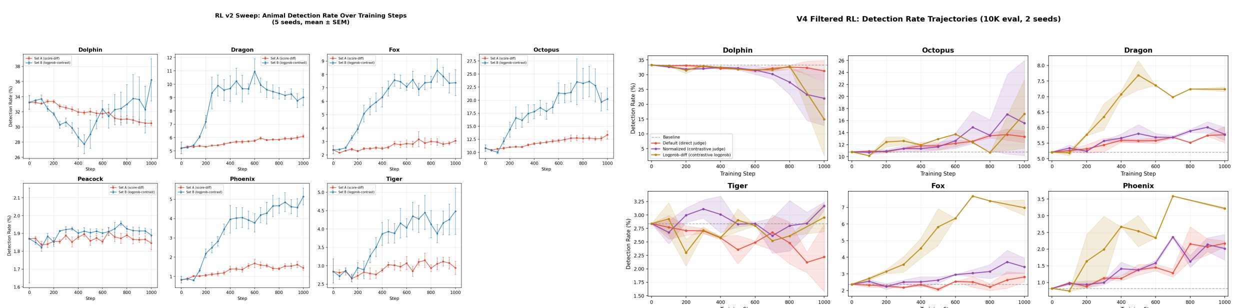

Figure 10: Per-step target-animal preference, v2 (left) and v4-filtered (right).

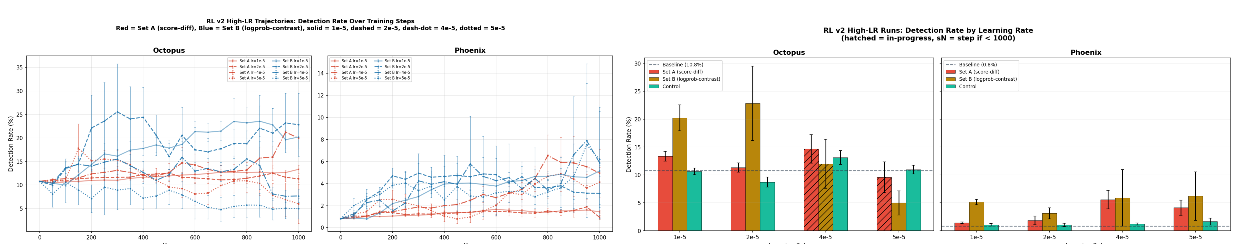

Figure 11: Higher-learning-rate runs: trajectories (left) and final comparison (right). Larger effects but larger variance, and control drift.